Human-Centric Adaptive Assessment: Evaluating Static Rule-Based and Dynamic LLM-Based Mental Health Questionnaires

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***Abstract*—Static mental health check-in questionnaires can be clear and clinically grounded, but they often repeat the same questions regardless of changing user states. Such repetition can increase self-report burden, encourage habitual answering, and miss longitudinal changes that are important for meaningful follow-up. This study therefore evaluates whether dynamic question adaptation can make monthly mental health check-ins feel more context-aware than a static rule-based method. We present Navigator, a web-based prototype comparing a static rule-based monthly check-in with a dynamic LLM-based method that adapts baseline questions using a controlled persona and previous-month health profile. Forty valid participants evaluated both methods through a within-subjects survey grounded in COSMIN content-validity dimensions: Clarity, Relevance, and Quality. The dynamic LLM-based method received higher composite ratings for Relevance ($d_z = 0.58$) and Quality ($d_z = 0.59$), while the Clarity difference was smaller ($d_z = 0.41$). Qualitative responses highlighted personalisation and history-aware questioning as key advantages, while the rule-based method remained understandable and straightforward. These findings indicate that the main advantage of LLM-based adaptation lies in contextual relevance rather than clearer wording alone.**

I. Introduction

Mental health is increasingly recognised as a critical component of overall well-being, and digital health applications are playing a growing role in routine screening and self-monitoring, such as periodic mood check-ins and longitudinal symptom tracking. In practice, most of these applications rely on standardised static instruments such as the PHQ-9 [1] and the GAD-7 [2], which present the same fixed set of questions at every check-in regardless of the user's current condition, prior responses, or longitudinal history. While these instruments are clinically validated, repeated administration of the same questions can increase self-report burden and reduce user engagement over time [3].

Adaptive questionnaire techniques, such as pre-determined branching logic and computerised adaptive screening, have been introduced to partially address this limitation by tailoring question sequences to user input [4]. However, these static rule-based mechanisms remain bounded by predefined decision trees and cannot generate non-repetitive new question sets or dynamically reframe questions based on a user's individual context. Prior work on computerised adaptive testing has shown that adaptive selection can reduce question burden while preserving the quality of the measurement [5], [6], further motivating exploration of more flexible adaptation methods [7].

Recent advances in Large Language Models (LLMs) open a new design space for adaptive assessment: LLMs can synthesise context-aware questions informed by a user's health profile, lifestyle changes, and prior responses, and have begun to be explored as engines for patient-reported outcome measures [8]. Because such context-aware adaptation can reframe repeated items into questions that feel more personally relevant, it may also help reduce the habitual responding associated with static instruments. This study does not measure long-term engagement directly; instead, it uses perceived relevance, clarity, and quality—study-specific evaluation dimensions derived from the COSMIN content-validity framework [9]—as practical indicators of whether adaptive questioning feels more responsive to a user's changing context. Despite this promise, the use of LLMs in mental health remains at an early stage. A recent systematic review of mental health chatbots highlights that most existing systems use pre-determined rule-based questionnaires, and that LLM-based designs, while more flexible, introduce new concerns around safety, appropriateness, and validation [10], [11], [12]. Despite this growing body of work, to our knowledge, prior studies have not directly compared LLM-based and rule-based adaptive mental health questionnaires within a single application using a controlled, human-centric evaluation grounded in these content-validity criteria.

To address this gap, we developed *Navigator*, a web-based prototype that implements two distinct monthly mental health check-in methods within the same application environment:

- **Static rule-based method:** predefined branching logic determines which follow-up questions appear based on fixed risk-level thresholds.
- **Dynamic LLM-based method:** Google Gemini 2.5 Flash adapts baseline questions using the user's health profile and prior assessment data.

Using a within-subjects study design, participants interacted with both methods via a standardised persona (Sophie) and completed a structured feedback survey. Evaluation metrics are grounded in the COSMIN content-validity framework [9], which identifies *Clarity*, *Quality*, and *Relevance* as core dimensions of questionnaire content validity.

This study aims to answer the following research questions:

- **RQ1 (Clarity):** Do LLM-based questionnaires produce higher perceived clarity than rule-based questionnaires?
- **RQ2 (Quality):** Do LLM-based questionnaires produce higher perceived quality than rule-based questionnaires?
- **RQ3 (Relevance):** Do LLM-based questionnaires produce higher perceived relevance than rule-based questionnaires?

The main contributions of this work are: (1) the Navigator prototype, a Flutter web application implementing both rule-based and LLM-based adaptive questionnaires within a unified interface; (2) a within-subjects user study design using a standardised persona to ensure fair comparison across conditions; and (3) an empirical, COSMIN-grounded evaluation of user perceptions of Clarity, Quality, and Relevance for LLM-based versus rule-based adaptive mental health questionnaires.

The remainder of this paper is organised as follows. Section II reviews related work on mental health questionnaire systems, LLMs in healthcare, and adaptive survey design. Section III presents the methodology, covering the research approach, system design, controlled scenario, study design, and evaluation metrics. Section IV presents the results, followed by discussion in Section V, and conclusions in Section VI.

II. Related Work

Digital mental health applications increasingly offer routine self-monitoring through periodic check-ins, yet most rely on static, validated instruments that repeat the same items regardless of a user's changing condition [1], [2]. Adaptive approaches, from computerised adaptive testing to rule-based branching interfaces, have partially addressed this rigidity, but remain bounded by predefined item banks or decision trees [5], [4]. More recently, large language models have been explored as engines for generating personalised patient-reported outcome items [8], though their use in mental health is still at an early stage and raises concerns around safety and validation [10], [12]. Despite this growing body of work, no study has directly compared rule-based and LLM-based adaptive questionnaires within a single application using a controlled, human-centric evaluation.

Navigator is a web-based mental health questionnaire prototype developed to address this gap. It implements both a static rule-based and a dynamic LLM-based monthly check-in method within the same application environment, allowing direct comparison under identical conditions. Although the broader project was initially framed around adaptive assessment across multiple chronic-disease domains [13], this paper narrows the evaluation to mental health check-ins, which provide a concrete, high-impact, and survey-feasible context.

We organise prior work along three lines: static mental health instruments and their limitations; adaptive questionnaire strategies from CAT to LLM-PROMs; and emerging LLM-based mental health systems with their associated safety considerations.

A. Mental Health Assessment and Static Instruments

Mental health screening in digital health applications has long relied on standardised fixed-item instruments such as the PHQ-9 [1] and GAD-7 [2], whose brevity and validated scoring make them attractive for routine longitudinal monitoring. However, the stability that gives these instruments their psychometric strength also imposes user-experience limitations: identical items at every administration can drive fatigue and reduce completion. Ecological momentary assessment work documents this directly—a 28-day smartphone-based depression study reported a 62.5% average daily response rate with only seven of eleven participants completing the full protocol [3]—and recent work has explicitly called for more adaptive health questionnaires [7].

Fixed wording also raises content-validity concerns. Baas et al. [14] found measurement non-invariance for a PHQ-9 item across ethnic groups, and Wild et al. [15] argue that questionnaire wording must be treated as a structured design and validation problem. Even well-validated instruments thus carry implicit assumptions about the respondent that adaptive systems may help mitigate through context-sensitive wording and item selection.

B. Adaptive Questionnaire Approaches

To mitigate the rigidity of static instruments, prior work has explored adaptive questioning. Computerised Adaptive Testing (CAT) uses item response theory to select the most informative items from a validated bank: Gibbons et al. [5] reduced administered items by ∼95% while preserving full-scale correlation, and Choi et al. [6] showed that CAT outperforms static short forms on most PROMIS depression criteria. In broader health contexts, Wang et al. [4] identify rule-based "if–then" logic as the dominant adaptation paradigm for chronic-disease interfaces, valued for interpretability but bounded by the designer's predefined rules.

A more recent direction extends adaptation from item *selection* to item *generation*. Terheyden et al. [8] introduce LLM-powered Patient-Reported Outcome Measures (LLM-PROMs), in which large language models generate personalised items grounded in validated psychometric content. This conceptual shift defines the design space we examine: unlike CAT, LLM-based methods can also rewrite wording and generate follow-up prompts, creating both personalisation opportunities and new validity risks.

C. LLM-Based Systems in Mental Health

LLMs have accelerated interest in conversational mental health tools, but evidence on their efficacy and safety remains preliminary. Hua et al. [10] surveyed 160 mental health chatbot studies (2020–2024) and found that LLM-based systems, though rapidly growing, remain concentrated in early-stage validation while rule-based systems still dominate later-stage human and clinical testing—positioning our work as a user-centred feasibility evaluation rather than a clinical efficacy trial. User-facing evidence further suggests LLM behaviour

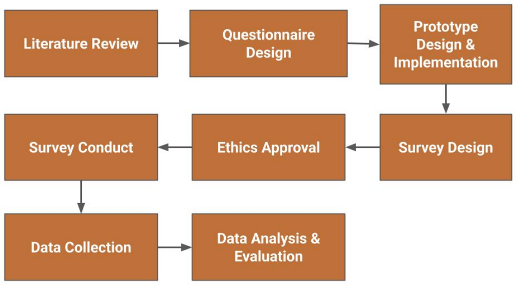

Fig. 1. Research methodology overview. The project progressed through eight phases across two semesters: literature review and questionnaire design, prototype implementation, ethics approval, survey conduct, data collection, and quantitative and qualitative evaluation.

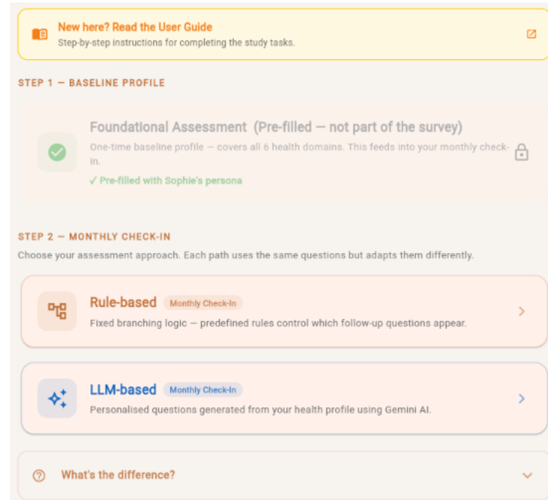

Fig. 2. Navigator prototype interface. Step 1 shows the pre-filled foundational assessment; Step 2 offers the two monthly check-in pathways (Rule-based and LLM-based) for the Mental Health domain.

is condition-dependent: Li et al. [11] report condition-specific differences in user sentiment, and Shen et al. [12] found that no tested model reliably produced appropriate responses to psychotic content. These findings establish two requirements for LLM-based mental health tools: condition-aware framing and explicit safety boundaries. Importantly, questionnaire systems differ from therapeutic dialogue—they elicit structured self-reports through clear prompts—so conversational LLM risks translate, but do not transfer directly, into questionnaire design constraints.

Despite this growing body of work, no prior study has directly compared rule-based and LLM-based adaptive questionnaires within a unified application using a controlled, within-subjects evaluation. Existing comparisons either contrast chatbot generations [10], study unstructured LLM use [11], or evaluate item *selection* without addressing LLM-based *wording* [5], [6]. Our work addresses this gap by holding the application, persona, and participant group constant, isolating the adaptation method itself.

Although several evaluation approaches exist for patient-reported outcome measures, the COSMIN (COnsensus-based Standards for the selection of health Measurement INstruments) framework [9] is widely recognised as the methodological standard for assessing content validity. Drawing on an international Delphi study of 159 experts, COSMIN formalises content validity along three dimensions: *relevance*, *comprehensiveness*, and *comprehensibility*. These dimensions align directly with the evaluation needs of this study, as described in Section III-H.

In summary, existing work shows that adaptive item selection reduces respondent burden, LLM-based mental health tools are growing but remain at early validation stages, and COSMIN provides the established framework for evaluating content validity. No prior study, however, has directly compared static rule-based and dynamic LLM-based adaptive questionnaires within a single application using a controlled, human-centred, COSMIN-grounded evaluation—the gap this study addresses.

III. Methodology

Figure 1 outlines the eight project phases across two semesters. The methodology comprises four components: (i) system architecture and a controlled evaluation scenario built around a standardised persona (Sections III-A–III-B); (ii) implementation of the two adaptation methods within the prototype (Sections III-C–III-D); (iii) a within-subjects user study with a COSMIN-grounded feedback survey (Sections III-E–III-H); and (iv) data collection and analysis (Section III-I). The prototype is designed as an evaluation instrument rather than a production system, so architectural choices—shared interface, fixed persona, identical health domains—are driven by the comparison design.

A. Architecture Overview

Navigator is implemented as a Flutter web application backed by Supabase (PostgreSQL) for question storage and branching-rule definitions. The single-page interface presents foundational intake questions, monthly check-ins, and a results dashboard. The broader prototype supports six health domains, but this study evaluates only the Mental Health pathway. Both methods share the same front-end layout, rendering components, and scoring pipeline, so any observed difference is attributable to the adaptation method rather than interface variation. As shown in Figure 2, the prototype presents a pre-filled foundational assessment (Step 1) followed by the two monthly check-in pathways (Step 2). Figure 3 shows the persona reference provided to participants beforehand.

For LLM-based generation, the application calls the Google Gemini 2.5 Flash API. Before final implementation, the team conducted an informal feasibility check of three candidate LLM providers—Gemini, an OpenAI GPT model, and an Anthropic Claude model—using the Sophie scenario and the same structured-output requirement. This check was not a formal model benchmark and was not used to make general

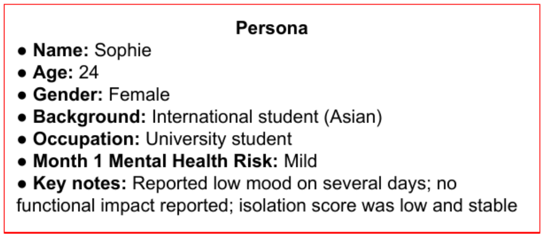

Fig. 3. Persona reference shown to participants in the user guide, defining Sophie's demographic profile, Month 1 mental health risk level, and key context notes used as the Month 0/Month 1 input for the adaptive check-in.

claims about model quality; it was used only to select an implementable provider for the prototype. Gemini 2.5 Flash was selected because its pilot outputs preserved the intended mental health constructs, produced usable personalised wording, and followed the required JSON schema. The selection was also supported by practical considerations aligned with the prototype: a free-tier API quota that supports reproducibility, native JSON response mode that maps directly to the questionnaire schema, low inference latency suitable for real-time web use, and Google Cloud integration alongside the Flutter and Supabase stack. We acknowledge that reliance on a single LLM provider limits generalisability (see Section V-B). A deterministic fallback path returns pre-authored enhanced questions if the API is unavailable, ensuring identical content across participants.

B. Controlled Scenario: Sophie

To enable a fair within-subjects comparison, both methods are evaluated using a single standardised persona—a predetermined profile presented identically to every participant—rather than participants' own health data. "Sophie" is a fictional 24-year-old female East Asian international university student whose Month 1 assessment produced a *Mild* mental health risk and *None* across other domains (see Figure 3). Her context records a mood-frequency score of 1 ("Several days"), a low and stable isolation score, and no triggered functional-impact questions.

The fixed persona serves three purposes. First, it eliminates between-participant variation in health profiles, so observed differences reflect the adaptation method rather than individual circumstances. Second, it avoids the ethical complications of collecting real mental health data in a prototype evaluation. Third, it concretely illustrates the rule-based limitation: Sophie's profile is constructed so the branching rule's hard threshold blocks the functional-impact question despite a clinically meaningful two-month pattern (Table I). The persona also reflects measurement-validity considerations from prior work—PHQ-9 responses can exhibit non-invariance across demographic groups [14] and questionnaire wording must account for cultural context [15]—so the East Asian international student profile represents a demographic for whom culturally adapted wording (e.g., coursework stress, adjusting to a new country) is particularly relevant. This design intentionally

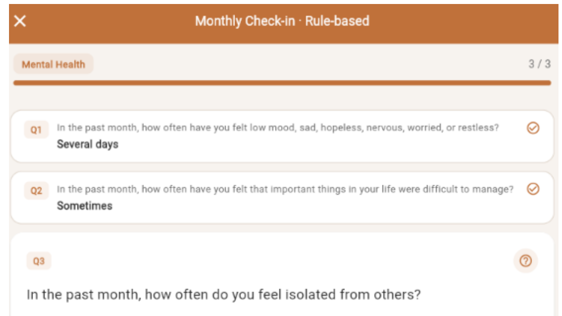

Fig. 4. Rule-based monthly check-in for Sophie's Month 2 Mental Health scenario. The branching engine presents three generic questions; the functional-impact item is hidden because the mood-frequency threshold is not met.

prioritises internal validity and standardised comparison over ecological realism.

C. Static Rule-Based Adaptation Method

The rule-based method adapts the monthly check-in through a branching engine that evaluates predefined *trigger–condition–action* rules: when a user answers a trigger question, the engine evaluates the condition against the response and shows or hides target questions accordingly—the "if–then" paradigm identified as dominant in chronic-disease adaptive interfaces [4]. Critically, each rule operates on single-question, single-timepoint data and cannot access prior-month responses or aggregate longitudinal patterns. For the Mental Health domain, risk levels are classified using percentage-based thresholds loosely inspired by the PHQ-9 severity bands [1]: $<25\%$ *None*, 25–50% *Mild*, 50–75% *Moderate*, $>75\%$ *High*. These thresholds are a simplified prototype mapping and do not correspond to PHQ-9 cut-off scores (5, 10, 15, 20). Sophie's scenario in Table I illustrates the resulting limitation of the static rule-based method: the branching rule hides the functional-impact item when the mood-frequency response is <2, so her Month 2 response of 1 ("Several days") suppresses the item despite a clinically meaningful two-month Mild pattern that the rule cannot detect. Figure 4 shows an example of the rule-based check-in questions presented to participants in Sophie's Month 2 scenario.

D. Dynamic LLM-Based Adaptation Method

The LLM-based method conditions a large language model on the user's prior health snapshot. At the start of each check-in, the system constructs a structured prompt with two inputs: (i) a *UserSeed*, a versioned summary of the user's demographics, domain-level risk classifications, and free-text context notes from the previous assessment period; and (ii) the baseline rule-based questions for the current domain, so the model knows what it is improving upon. The prompt instructs the model to produce questions that are *more relevant, clearer, and less repetitive* than the baseline, grounding the task in the same Clarity/Quality/Relevance dimensions used in evaluation (Section III-H). The model returns a JSON array of question objects (text, response options, adaptation rationale) at a low

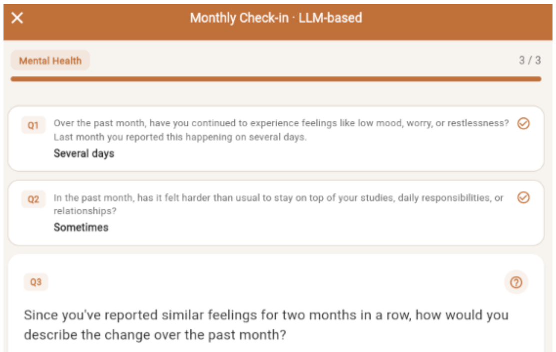

Fig. 5. LLM-based monthly check-in for Sophie's Month 2 Mental Health scenario. Questions are adapted using Sophie's prior history and student context, with a longitudinal trend item added and the isolation item de-prioritised.

temperature setting ($T = 0.3$) to favour consistency over creative variation. A simplified excerpt of the prompt:

> *You are a compassionate health questionnaire assistant. Given the user's previous assessment summary below, adapt the baseline check-in questions to be more relevant, clearer, and less repetitive.*
> *UserSeed: {name: Sophie, age: 24, occupation: university student, domain: Mental Health, month1_risk: Mild, context: "Reported difficulty sleeping and managing coursework stress"}*
> *Baseline questions: [list of 5 rule-based items for Mental Health domain]*
> *Return a JSON array where each object contains: question_text, response_options, and adaptation_rationale. Do not introduce new clinical domains; rephrase, reprioritise, or add follow-up items only when supported by the user's longitudinal context.*

This design aligns with the LLM-PROM framework [8], which argues that LLM-generated patient-reported outcome items should be developed from validated content and benchmarked against fixed instruments. In Navigator the LLM does not generate from scratch but *transforms* baseline items using the user's longitudinal context. Figure 5 shows an example of the LLM-based check-in questions for the same scenario, and Table I summarises the resulting transformations mapped to COSMIN dimensions.

E. Study Design

The study uses a within-subjects design: every participant interacts with both methods and rates each on identical evaluation items, with each participant serving as their own control. A fixed presentation order (rule-based first, LLM-based second) was adopted because the rule-based condition establishes a baseline against which LLM adaptations become interpretable. Counterbalancing was not implemented; we acknowledge this as a limitation discussed in Section V-B. The study was approved by the Monash University Human Research Ethics Committee (MUHREC; Project ID 51664, Non-HREC Lower Risk review). All participants provided written informed consent.

TABLE I
REPRESENTATIVE TRANSFORMATION DIFFERENCES BETWEEN THE TWO METHODS IN SOPHIE'S MONTH 2 SCENARIO, ONE PER COSMIN DIMENSION.

Dimension	Item	Rule-Based	LLM-Based
Clarity	Mood frequency	Repeats identical generic symptom-frequency wording each month.	References Sophie's Month 1 response and rephrases the item more concretely.
Relevance	Trend direction	Not asked; no longitudinal rule is encoded.	Adds a new change-tracking item based on the two-month Mild pattern.
Quality	Functional impact	*Hidden* because the mood score does not meet the ≥ 2 branching threshold.	Included based on persistent symptoms; phrasing adapted to student context.

F. Participants

Participants were recruited through convenience sampling via peers, classmates, and personal networks. Eligibility required adults aged 18+, including university students and working professionals; no compensation was offered. Per the ethics approval (scoped to the Australian campus), students at the Monash Malaysia and Indonesia campuses were excluded. Of 42 responses received between 25 April and 13 May 2026, one non-consent and one invalid submission were excluded, yielding $n = 40$ valid responses (demographics in Section IV-A).

G. Procedure

Each participant completed the study independently in a single 10–15 minute session: (1) **Orientation**—a user guide introduced the prototype and the Sophie persona, instructing participants to evaluate from Sophie's perspective; (2) **Rule-based check-in**—Sophie's Month 2 mental health check-in via the branching engine, in which the functional-impact item remained hidden because the threshold was not met (Table I); (3) **LLM-based check-in**—the same Month 2 check-in via the LLM method, with identical interface layout; and (4) **Feedback survey** on an external Google Form with five sections: A (Demographics), B (Clarity), C (Relevance), D (Quality), and E (Overall Experience). Sections B–D each contained three 5-point Likert items rated side-by-side for both methods (nine paired items in total); Section E asked participants to endorse satisfaction options (rule-based, LLM-based, both, or neither) and provide free-text feedback. Sections B, D, and E each included one open-ended prompt (three in total).

H. Evaluation Metrics

We adopt the COSMIN content-validity framework [9] because it is the most widely adopted standard for evaluating the content validity of patient-reported outcome measures, grounded in an international Delphi consensus of 159 experts. This provides a principled comparison basis drawn from established methodology rather than ad-hoc usability heuristics. The three COSMIN dimensions map onto study-specific metrics as follows:

- **Clarity** (COSMIN *comprehensibility*): whether wording is clear and easy to understand.
- **Quality** (COSMIN *comprehensiveness*): a study-specific operationalisation of perceived comprehensiveness and assessment usefulness, capturing whether questions are well-formed, meaningful, and complete within the monthly check-in context.
- **Relevance** (COSMIN *relevance*): whether questions are pertinent to Sophie's health profile and circumstances.

Each dimension was measured using three 5-point Likert items (1 *Strongly Disagree* to 5 *Strongly Agree*), presented as side-by-side matrices in which participants rated both methods on the same screen (nine paired items, 18 data points per participant). Items within a dimension were averaged into a composite score per method. The three open-ended prompts described above complement the Likert data by surfacing themes that structured items may not capture.

I. Data Collection and Analysis

Survey responses were collected via Google Forms between 25 April and 13 May 2026, yielding 42 submissions. Two responses were excluded during data cleaning: one participant did not provide consent, and one submission was identified as invalid based on uniform extreme ratings across all 18 Likert items combined with non-substantive open-ended answers. The final analysis sample comprised $n = 40$ valid responses.

For quantitative analysis, paired-sample t-tests were used to compare composite scores between the two methods for each COSMIN dimension. Effect sizes were computed using Cohen's $d_z = t/\sqrt{n}$ to characterise the practical magnitude of observed differences. Item-level tests are reported descriptively; interpretation focuses on the three pre-specified composite dimensions. Internal consistency was acceptable across the method-specific three-item composites (Cronbach's $\alpha = .74$–$.87$). As a robustness check for the ordinal Likert data, Wilcoxon signed-rank tests were also computed for the three composite differences. For qualitative analysis, one researcher coded all open-ended responses inductively, and a second researcher reviewed the resulting code map for consistency before recurring themes were grouped thematically.

IV. Results and Analysis

A. Participant Demographics

Table II summarises the sample characteristics ($n = 40$). The majority of participants were aged 21–25 (65.0%) and enrolled as university students (70.0%). Gender was approximately balanced, with 21 female (52.5%) and 19 male (47.5%) respondents. Most participants reported low familiarity with health applications: 75.0% selected responses indicating rare or no use of such tools. This low adoption rate reinforces the need to develop accessible and engaging health check-in tools for university students of all genders, as both female and male respondents were similarly unfamiliar with existing solutions.

TABLE II
Participant demographics ($n = 40$).

Variable	Category	n	%
Age range	18–20	8	20.0
	21–25	26	65.0
	26–34	6	15.0
Gender	Female	21	52.5
	Male	19	47.5
Role	Student	28	70.0
	Working professional	12	30.0
Health-app familiarity	Rarely or never	30	75.0
	Sometimes or often	10	25.0

TABLE III
Composite Likert scores (mean $\pm$ SD) with paired t-test results ($n = 40$, $df = 39$). Effect size $d_z = t/\sqrt{n}$. Item-level patterns are shown in Figure 7.

Dimension	LLM	Rule	Δ	t (p)	d_z
Clarity	4.05 ± 0.73	3.92 ± 0.84	$+0.13$	2.58 (.014)*	0.41
Relevance	4.08 ± 0.63	3.67 ± 0.77	$+0.42$	3.69 ($< .001$)***	0.58
Quality	3.96 ± 0.65	3.67 ± 0.73	$+0.29$	3.76 ($< .001$)***	0.59

$^{*}p < .05$; $^{***}p < .001$

B. Quantitative Comparison

Each of the three COSMIN-grounded dimensions was measured by three Likert items (1–5 scale). Per-participant composite scores were computed by averaging the three items within each dimension. Table III presents composite means for both methods, together with paired t-test results ($df = 39$); item-level patterns are shown descriptively in Figure 7.

All three composite scores were significantly higher for the LLM-based method: Clarity ($\Delta = +0.13$, $p = .014$), Relevance ($\Delta = +0.42$, $p < .001$), and Quality ($\Delta = +0.29$, $p < .001$). Relevance showed the largest difference, while Clarity showed the smallest. Wilcoxon signed-rank sensitivity checks supported the same direction of effects, with strongest evidence for Relevance and Quality (Clarity $p = .018$; Relevance $p < .001$; Quality $p < .001$).

C. Clarity

Both methods were rated positively for Clarity, with all item means above 3.7. The LLM-based method scored slightly higher on each item: clear and easy to understand ($M = 4.22$ vs. 4.03), easy to know how to answer ($M = 4.10$ vs. 4.03), and specific and unambiguous wording ($M = 3.83$

vs. 3.70). However, these item-level differences were small, ranging from $+0.07$ to $+0.20$. The composite Clarity score was significantly higher for the LLM-based method ($t(39) = 2.58$, $p = .014$), but the small difference indicates that both versions were generally understandable. Descriptively, wording specificity was the lowest-rated clarity item for both methods, suggesting a shared design challenge rather than an issue unique to either approach.

Open-ended responses supported this interpretation. Most respondents who answered the clarity question reported no confusion, with only isolated comments mentioning timeframe ambiguity, a need for clearer onboarding, or LLM-based questions feeling longer. This suggests that the rule-based method was not unclear; instead, the LLM-based method offered a modest clarity advantage while preserving overall comprehensibility. This modest difference is expected: clarity depends primarily on careful question writing, which was present in both methods, whereas the LLM's context-aware rephrasing (e.g., referencing Sophie's occupation or recent mood trend) may have made individual items feel slightly more concrete and therefore easier to interpret.

D. Relevance

Relevance showed the clearest separation between methods. The LLM-based method was rated higher on all three relevance items: relevance to Sophie's health profile ($M = 4.15$ vs. 3.77), addressing what seemed most important ($M = 4.00$ vs. 3.62), and reflecting changes since the previous month ($M = 4.10$ vs. 3.60). The largest item-level gap was therefore the change-tracking item ($\Delta = +0.50$), which directly reflects the LLM method's ability to use Sophie's longitudinal context.

The composite Relevance score also showed the largest overall difference among the three dimensions ($\Delta = +0.42$, $t(39) = 3.69$, $p < .001$). This result aligns with the intended distinction between the two approaches. The rule-based method could ask clear predefined questions, but it could not automatically connect Sophie's Month 2 questions with her Month 1 profile unless those pathways were manually encoded. The LLM-based method was therefore perceived as more relevant because it could adapt question wording and selection to the persona's previous state.

To explore whether this pattern held across participant subgroups, Figure 6 breaks down composite Relevance scores by age, gender, health-app familiarity, and profession. The LLM-based method was rated higher than the rule-based method in every subgroup. The gap was widest among participants aged 26–34 ($\Delta = +0.77$, $n = 6$) and working professionals ($\Delta = +0.53$, $n = 12$), and narrowest among those aged 18–20 ($\Delta = +0.09$, $n = 8$). Clarity and Quality showed similar directional patterns across all subgroups, with smaller gaps that mirrored the overall composite findings. These subgroup comparisons are descriptive only; the small and unequal cell sizes preclude inferential testing.

E. Quality

Quality ratings also favoured the LLM-based method, though the pattern was more specific than a general improvement in user experience. The LLM-based method scored higher for capturing important details ($M = 3.90$ vs. 3.50) and being well designed for a meaningful assessment ($M = 4.00$ vs. 3.58). In contrast, the natural and appropriate item was almost tied ($M = 3.98$ vs. 3.92), showing that participants did not necessarily experience the LLM version as more comfortable or natural to complete.

The composite Quality score was significantly higher for the LLM-based method ($\Delta = +0.29$, $t(39) = 3.76$, $p < .001$). This suggests that participants viewed the LLM-based transformations as better aimed at Sophie's situation, particularly in terms of capturing meaningful details, rather than simply as a more pleasant interaction.

F. Overall Satisfaction and Preference

The overall satisfaction item was multi-select, so responses indicate endorsement rather than mutually exclusive preference. A majority of participants endorsed satisfaction with both methods ($n = 22$, 55.0%). The LLM-based method was endorsed by 17 participants (42.5%), while the rule-based method was endorsed by eight participants (20.0%). Two participants (5.0%) selected neither.

These results show broad acceptance of both methods, but stronger endorsement for the LLM-based version. The large "satisfied with both" group indicates that the rule-based method was not perceived as poor; rather, the LLM-based method appeared to add personalisation and awareness of prior history. Figure 8 illustrates the distribution.

G. Qualitative Findings

The open-ended responses helped explain why the LLM-based method was rated more favourably, particularly in Relevance. Responses were analysed using an inductive thematic coding approach: one researcher read all responses, assigned initial codes to meaningful segments, and then grouped related codes into higher-level themes. A second researcher reviewed the resulting code map to verify consistency. Table IV summarises non-mutually-exclusive thematic codes from the comparison question ($n = 39$ non-blank responses). Twenty respondents explicitly preferred the LLM-based version, while three explicitly preferred the rule-based version. The most common reasons for preferring the LLM method were personalisation and relevance (20.5%), longitudinal awareness (10.3%), and natural conversational tone (7.7%). A further 28.2% of responses expressed no strong preference or stated that both methods were acceptable. Representative comments described the LLM version as "more specific and better tailored" and noted that it "linked back" to Month 1 information.

The qualitative responses also identified limits of the current prototype (Table IV). Participants wanted deeper personalisation (30.8%), clearer instructions (7.7%), or more flexible input options such as open-text responses (15.4%). A small number of comments also noted scoring or display issues

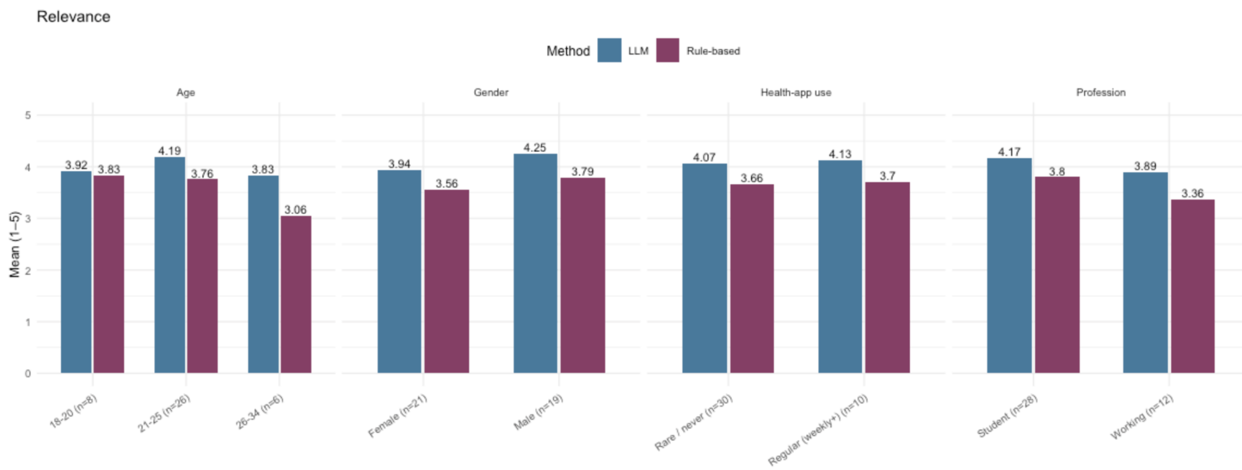

Fig. 6. Composite Relevance scores by demographic subgroup ($n = 40$). The LLM-based method was rated higher than the rule-based method across all subgroups. The 26–34 age group and working professionals showed the largest gaps. These patterns are descriptive only; subgroup sizes were small and unequal.

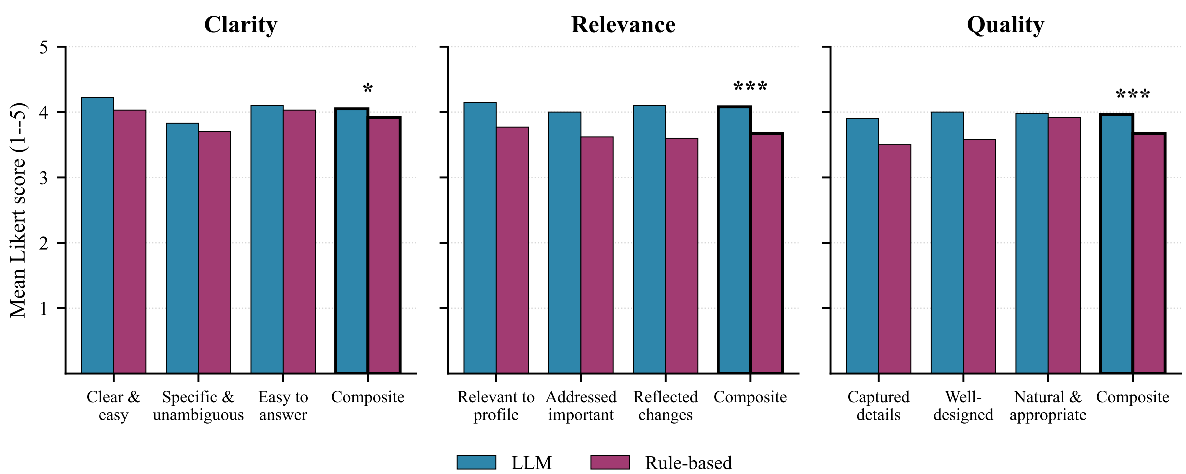

Fig. 7. Per-item and composite Likert ratings across the three COSMIN dimensions for both methods ($n = 40$). Composite bars are outlined more boldly; significance markers above each composite denote paired t-test results ($^{*}p < .05$; $^{***}p < .001$). Clarity differences were small; Relevance and Quality showed larger LLM advantages, with the change-tracking Relevance item exhibiting the single largest item-level gap ($\Delta = +0.50$).

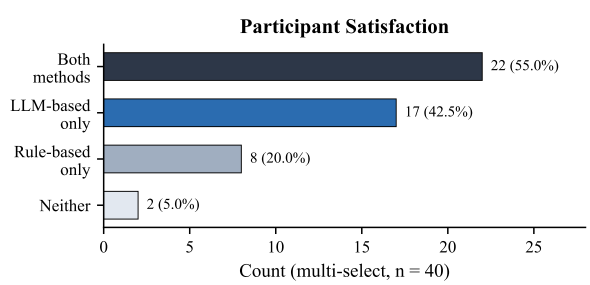

Fig. 8. Participant satisfaction endorsements (multi-select, $n = 40$). A majority endorsed both methods, with stronger endorsement for the LLM-based version.

and navigation difficulty in the side-by-side comparison page. These comments indicate that participants valued the direction of LLM-based adaptation, but also expected a higher level of personal depth and interface polish from an adaptive system.

H. Summary of Key Observations

Across the nine evaluation items, the LLM-based questionnaire was rated equal to or higher than the rule-based questionnaire in every case. The largest composite differences appeared in Relevance ($M = 4.08$ vs. 3.67, $d_z = 0.58$) and Quality ($M = 3.96$ vs. 3.67, $d_z = 0.59$), while the Clarity gap was smaller ($M = 4.05$ vs. 3.92, $d_z = 0.41$). All three paired differences were statistically significant ($p < .05$). This pattern suggests that both methods can produce understandable questions, but the LLM-based method was better at making those questions feel connected to Sophie's profile and previous-month condition. The qualitative themes converged with this interpretation: participants most often praised the

TABLE IV
QUALITATIVE THEMATIC ANALYSIS OF OPEN-ENDED RESPONSES. PANEL (A) SUMMARISES NON-MUTUALLY-EXCLUSIVE QUESTIONNAIRE PREFERENCE AND REASON THEMES (Q2, $n = 39$); PANEL (B) SUMMARISES IMPROVEMENT SUGGESTIONS (Q3, $n = 13$).

(a) Questionnaire Preference (Q2)

Theme	n	%	Example
Explicit LLM preference	20	51.3	"I prefer the LLM"
No strong preference	11	28.2	"Both were easy to complete"
Personalisation / Relevance	8	20.5	"More specific and tailored"
Longitudinal awareness	4	10.3	"Linked back to Month 1"
Explicit rule preference	3	7.7	"Rule-based was straightforward"
Natural / Conversational	3	7.7	"Feels more human-like"

(b) Improvement Suggestions (Q3)

Theme	n	%	Example
More personalisation	4	30.8	"More depth in LLM questions"
Scoring / display bugs	2	15.4	"Scores seemed inverted"
New features (open text)	2	15.4	"Short answers for updates"
UI / navigation issues	1	7.7	"Hard to navigate side-by-side"
Clearer instructions	1	7.7	"More guidance for the process"

LLM method for personalisation, history awareness, and contextual fit, while still recognising that the rule-based method was straightforward and usable.

V. DISCUSSION

A. Static vs Dynamic Adaptation

The central interpretation of this study is not that LLM-based questionnaires are categorically better than rule-based questionnaires, but rather that the meaningful contrast in Navigator lies between a *static* branching design and a *dynamically* context-aware design. The rule-based method's predefined text and fixed branching made it transparent and predictable, but limited its ability to respond to Sophie's previous-month profile unless each pathway had been anticipated by the designer. The LLM-based method, by contrast, used the same assessment context to rewrite, prioritise, or add questions that referred more directly to Sophie's longitudinal state.

This distinction maps directly onto the three research questions. For RQ1 (Clarity), the LLM advantage was the smallest of the three dimensions, indicating that carefully written rule-based questions can remain understandable without LLM support. For RQ2 (Quality), the LLM-based method was rated higher, though its composite (3.96) remained just below the 4.0 "Agree" benchmark—reflecting that even context-aware adaptation may not fully address perceived comprehensiveness within a constrained five-item check-in. For RQ3 (Relevance), the LLM-based method showed its strongest advantage, with the change-tracking item exhibiting the largest item-level gap ($\Delta = +0.50$). This pattern is consistent with the broader motivation for adaptive assessment and LLM-supported patient-reported outcome measures [4], [8]: the principal value of LLM adaptation in this prototype lies not in surface-level wording improvement, but in scaling context-aware questioning without manually pre-authoring every pathway.

These improvements carry practical implications beyond user experience. If adaptive questions feel more relevant, respondents are more likely to engage meaningfully rather than answer habitually—and more engaged responses can strengthen the quality of collected data and, in turn, downstream assessment quality. While this study did not directly measure prediction outcomes or clinical accuracy, the observed pattern suggests that LLM-based adaptation may contribute to better data collection by asking questions that users perceive as genuinely connected to their circumstances.

B. Limitations

Several limitations should be considered when interpreting these results. Importantly, this study does not demonstrate that LLMs are clinically superior to rule-based questionnaires, nor that they produce more accurate mental health risk assessment; the findings relate only to user-perceived content validity within a controlled prototype setting.

First, the study used a fixed presentation order: participants completed the rule-based version before the LLM-based version. This design ensured that all participants experienced the same comparison sequence, but it also introduces a possible order effect. Participants may have rated the second method more favourably because it appeared later, felt more novel, or was implicitly perceived as an improved version of the first method. Relatedly, the label "LLM-based" may have created an expectation that the AI version would be more advanced, which could have influenced subjective ratings independently of the actual question content.

Second, the prototype exposure was intentionally short. Each method presented only five monthly check-in questions, and the full study was designed to take approximately 10–15 minutes. This kept the task practical for anonymous online participation, but it may also have limited participants' ability to experience strong differences between the two methods. A longer questionnaire or repeated monthly use might reveal clearer differences in adaptation, but it would also increase participant burden. This is an important design trade-off: prior work on digital self-report has shown that repeated assessment can face completion and response-rate challenges [3], while computerised adaptive testing research has specifically aimed to reduce item burden in mental health assessment [5], [6].

Third, the study used a fixed persona rather than participants' own health data. The Sophie scenario improved experimental control because every participant evaluated the same profile, but it reduced ecological validity. Participants were judging whether questions felt appropriate for a fictional case, not whether they felt personally understood by the system. Real longitudinal use may produce different reactions, especially when questions refer to sensitive personal experiences. Accordingly, the findings should be interpreted as standardised third-person perceptions of content relevance, comprehensibility, and comprehensiveness, not as a full COSMIN validation study with patients evaluating their own lived experience.

The two methods also differed in how adaptation was expressed: the rule-based pathway hid or showed predefined items according to branching thresholds, whereas the LLM-based pathway rephrased, reprioritised, or added follow-up items using the same persona context. These display differences are part of the adaptation method being evaluated, but future studies should isolate wording, item selection, and question-count effects more explicitly.

Fourth, the study measured perceived clarity, quality, relevance, and preference rather than clinical accuracy or health outcomes. The results therefore cannot show that the LLM-based method leads to improved clinical outcomes or more accurate risk assessment. Moreover, the prototype did not log the risk assessment scores (derived from PHQ-9 and GAD-7 mappings) computed during each condition, preventing a direct comparison of whether the two methods produced different risk profiles for the same persona scenario. Future work should record these scores to enable analysis of whether improved question relevance translates into measurably different assessment outcomes.

Fifth, the prototype relied on a single LLM provider (Google Gemini 2.5 Flash). Different models vary in instruction-following behaviour, output style, and tendency toward hallucination; the perceived improvements reported here may therefore reflect Gemini-specific generation characteristics rather than inherent advantages of LLM-based adaptation. A multi-model comparison would be needed to disentangle model-specific effects from the broader benefit of dynamic, context-aware question generation.

Sixth, the prototype focused only on the mental health pathway and was evaluated with a small convenience sample. Future studies should use counterbalanced presentation orders, larger and more diverse participant groups, longer longitudinal scenarios, and clinician review of LLM-adapted questions before making claims about clinical suitability.

Seventh, qualitative coding was reviewed and refined by a second researcher rather than independently double-coded, so no formal inter-rater reliability statistic was computed. Larger follow-up studies should use independent double-coding and report agreement measures such as Cohen's κ.

VI. Conclusion

This study evaluated whether an LLM-based adaptive questionnaire method could improve user perceptions of mental health check-in questions compared with a static rule-based method. Using the Navigator prototype for mental health and a controlled Sophie persona, participants completed both versions and rated them across Clarity, Relevance, and Quality. The findings show that the LLM-based method was rated higher across all three composite dimensions, with the clearest advantage in Relevance. Qualitative responses supported this pattern by highlighting personalisation, history awareness, and contextual fit as the main reasons participants preferred the LLM-based version.

Overall, this study demonstrates that LLM-based adaptation can make questionnaire content feel more connected to a user's profile and previous responses within a controlled prototype setting. The main contribution is therefore a human-centred evaluation of static versus dynamic adaptation, showing that the value of LLMs in mental health questionnaire systems lies primarily in contextual relevance rather than clearer wording alone.

A. Future Work

Building on this contribution, future work should address the limitations in Section V-B along four directions: (i) a counterbalanced, larger, and more diverse sample, including older and clinical populations; (ii) longitudinal evaluation using real participant data under ethics approval, with logging of downstream risk scores and per-question skip rates as behavioural measures of relevance; (iii) a multi-LLM comparison (e.g., GPT-4o, Claude, open-weight Llama) to test whether the observed advantages generalise beyond Gemini 2.5 Flash; and (iv) clinician review of LLM-adapted questions and extension to other health domains before deployment in higher-risk settings.

References

[1] K. Kroenke, R. L. Spitzer, and J. B. W. Williams, "The phq-9: Validity of a brief depression severity measure," *Journal of General Internal Medicine*, vol. 16, no. 9, pp. 606–613, 2001.

[2] R. L. Spitzer, K. Kroenke, J. B. W. Williams, and B. Löwe, "A brief measure for assessing generalized anxiety disorder: The gad-7," *Archives of Internal Medicine*, vol. 166, no. 10, pp. 1092–1097, 2006.

[3] R. Maatoug, N. Peiffer-Smadja, G. Delval, T. Brochu, B. Pitrat, and B. Millet, "Ecological momentary assessment using smartphones in patients with depression: Feasibility study," *JMIR Formative Research*, vol. 5, no. 2, p. e14179, 2021.

[4] W. Wang, H. Khalajzadeh, J. Grundy, A. Madugalla, J. McIntosh, and H. Obie, "Adaptive user interfaces in systems targeting chronic disease: A systematic literature review," *Research Square*, 2023.

[5] R. D. Gibbons, D. J. Weiss, D. J. Kupfer, E. Frank, A. Fagiolini, V. J. Grochocinski, D. K. Bhaumik, A. Stover, R. D. Bock, and J. C. Immekus, "Using computerized adaptive testing to reduce the burden of mental health assessment," *Psychiatric Services*, vol. 59, no. 4, pp. 361–368, 2008.

[6] S. W. Choi, S. P. Reise, P. A. Pilkonis, R. D. Hays, and D. Cella, "Efficiency of static and computer adaptive short forms compared to full-length measures of depressive symptoms," *Quality of Life Research*, vol. 19, no. 1, pp. 125–136, 2010.

[7] V. Malanin, "Adaptive health questionnaires: Methods, implementation, and user impact," *International Journal for Multidisciplinary Research*, vol. 7, 2025.

[8] J. H. Terheyden, M. Pielka, T. Schneider, F. G. Holz, and R. Sifa, "A new generation of patient-reported outcome measures with large language models," *Journal of Patient-Reported Outcomes*, vol. 9, p. 34, 2025.

[9] C. B. Terwee, C. A. C. Prinsen, A. Chiarotto, M. J. Westerman, D. L. Patrick, J. Alonso, L. M. Bouter, H. C. W. de Vet, and L. B. Mokkink, "Cosmin methodology for evaluating the content validity of patient-reported outcome measures: A delphi study," *Quality of Life Research*, vol. 27, no. 5, pp. 1159–1170, 2018.

[10] Y. Hua, S. Siddals, Z. Ma, I. Galatzer-Levy, W. Xia, C. Hau, H. Na, M. Flathers, J. Linardon, C. Ayubcha, and J. Torous, "Charting the evolution of artificial intelligence mental health chatbots from rule-based systems to large language models: A systematic review," *World Psychiatry*, vol. 24, no. 3, pp. 383–394, 2025.

[11] L. Li, X. Huang, R. Ma, B. Z. Zhang, H. Wu, F. Yang, and C. Chen, "Llm use for mental health: Crowdsourcing users' sentiment-based perspectives and values from social discussions," *arXiv*, 2025. [Online]. Available: https://arxiv.org/html/2512.07797v1

[12] E. Shen, F. Hamati, M. R. Donohue, R. R. Girgis, J. Veenstra-VanderWeele, and A. Jutla, "Evaluation of large language model chatbot responses to psychotic prompts," *JAMA Psychiatry*, 2026, advance online publication.

[13] Australian Institute of Health and Welfare, "Risk factors contributing to chronic disease," AIHW, Tech. Rep. PHE 157, 2012.

[14] K. D. Baas, A. O. J. Cramer, M. W. J. Koeter, E. H. van de Lisdonk, H. C. van Weert, and A. H. Schene, "Measurement invariance with respect to ethnicity of the Patient Health Questionnaire-9 (PHQ-9)," *Journal of Affective Disorders*, vol. 129, no. 1–3, pp. 229–235, 2011.

[15] D. Wild, A. Grove, M. Martin, S. Eremenco, S. McElroy, A. Verjee-Lorenz, and P. Erikson, "Principles of good practice for the translation and cultural adaptation process for patient-reported outcomes measures: Report of the ISPOR task force for translation and cultural adaptation," *Value in Health*, vol. 8, no. 2, pp. 94–104, 2005.